Manuscripts considered — aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun

Master inventory: every paper reviewed in this case

Generated 2026-08-12

Manuscripts considered — aml-mds-related-rr-tp53-aberrant-hla-p

Master inventory: 15 clinical (13 included, 2 considered & excluded) + 16 pre-clinical (13 included, 3 considered & excluded) + 20 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Gyurkocza/Giralt (2025) — J Clin Oncol

Reference:PMID 39298738 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Iomab-B (131I-apamistamab) anti-CD45 radioimmunotherapy conditioning
- **Indication / model:** Active relapsed/refractory AML, age >=55, planned allo-HCT (2L+); Active R/R AML age >=55 fit for allo-HCT; ITT included crossover (52% of conventional-care arm received 131I-apamistamab). Enrolled patients heading to a first allo-HCT.
- **Design:** Randomized open-label phase 3 (SIERRA)
- **n:** 153
- **Effect size:** 17.1 % — Durable complete remission (>=180 days)
- **Variance:** 95% CI 9.4–27.5; p=<0.0001
- **Toxicities:** Any treatment-related AE G>=3 (n=76, 59.7%)
- **Case fit:** partial
- **Notes:** Randomized evidence that CD45-targeted radioimmunotherapy can carry active R/R AML into allo-HCT, speaking to conditioning a second transplant with an 85%-blast marrow. SIERRA enrolled patients going to a first transplant, so prior-HCT eligibility is the axis to check. OS was flat because most control patients crossed over.

—/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** excluded — No peer-reviewed clinical efficacy data to extract yet: the solid-tumor PYNNACLE study (NCT04585750) and the myeloid rezatapopt + azacitidine trial (NCT06616636) are ongoing with results limited to conference abstracts. Also Y220C-contingent, and the patient's TP53 status and specific variant are unresolved.

- **Intervention:** Rezatapopt (PC14586, Y220C-selective p53 reactivator)
- **Notes:** Excluded: No peer-reviewed clinical efficacy data to extract yet: the solid-tumor PYNNACLE study (NCT04585750) and the myeloid rezatapopt + azacitidine trial (NCT06616636) are ongoing with results limited to conference abstracts. Also Y220C-contingent, and the patient's TP53 status and specific variant are unresolved. — The Y220C-selective reactivator is the actionable TP53 handle if a Y220C variant is confirmed; retained for the audit trail with no manuscript to source effect sizes or toxicities from.

Carter/Andreeff (2025) — Blood

Reference:PMID 40608889 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) in TP53-Y220C AML
- **Indication / model:** TP53-Y220C AML cell lines and stem/progenitor cells; TP53-Y220C AML xenograft mice
- **Design:** PC14586 (rezatapopt) binds the pocket created by the Y220C substitution and refolds p53 toward a

wild-type conformation that reactivates transcriptional targets; in AML, MDM2/XPO1 feedback plus loss of a p53-BCL-2/BAX interaction blunt apoptosis, so BCL-2 inhibition is needed to turn reactivation into cell death.
- **Effect size:** strong — Rezatapopt restored wild-type p53 conformation and target-gene activity in TP53-Y220C AML but induced little apoptosis alone. Adding venetoclax produced heavy killing of AML cells and stem/progenitors in vitro and prolonged survival of TP53-Y220C AML xenografts.
- **Variance:** vs rezatapopt alone; venetoclax alone; MDM2/XPO1-inhibitor arms; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Single-agent rezatapopt was largely cytostatic in AML; the survival benefit needed venetoclax, which this patient's disease has already progressed through in FLAG-IDA/venetoclax reinduction. PMV-affiliated co-authors; contingent on confirming a Y220C variant. Underpins NCT06616636.

Puzio-Kuter/Poyurovsky (2025) — Cancer Discov

Reference:PMID 39945593 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) Y220C reactivation POC
- **Indication / model:** TP53-Y220C human tumor cell lines and xenografts (solid tumors); biochemical, structural, and transcriptomic assays
- **Design:** Rezatapopt corrects the Y220C-mutant p53 conformation so it binds DNA and drives its target-gene program, restoring antiproliferative p53 activity, characterized biochemically, structurally, and by in-vitro and in-vivo transcriptomics.
- **Effect size:** strong — Rezatapopt recovered multiple arms of the wild-type p53 program in Y220C models and gave potent antitumor activity in preclinical xenografts as a single agent and with immunotherapy. This is the foundational proof-of-concept dataset for the clinical compound.
- **Variance:** vs vehicle; TP53-WT and non-Y220C mutant lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Solid-tumor models, so read-across to this patient's myeloid disease rests on the shared Y220C mechanism rather than tumor match. PMV Pharmaceuticals study.

Vu/Dumble (2025) — ACS Med Chem Lett

Reference:PMID 39811143 · **Type:** preclinical · **Inclusion:** excluded — Medicinal-chemistry discovery and structure report for rezatapopt with Y220C xenograft regression; the fuller preclinical efficacy characterization is captured by Puzio-Kuter 2025 and the AML-specific biology by Carter 2025.

- **Intervention:** Rezatapopt (PC14586) discovery / medicinal-chemistry report
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Medicinal-chemistry discovery and structure report for rezatapopt with Y220C xenograft regression; the fuller preclinical efficacy characterization is captured by Puzio-Kuter 2025 and the AML-specific biology by Carter 2025.

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CBX-250 (TCR-mimetic bispecific T-cell engager targeting CG1/HLA-A*02:01)

- **Indication / model:** R/R AML / high-risk MDS / CMML / CML (CROSSCHECK-001, first-in-human) (2L+); HLA-A *02:01 required; target is CG1 (Cathepsin-G-derived peptide) presented on HLA-A*02:01
- **Design:** Open-label phase 1 first-in-human dose-escalation, single-agent, subcutaneous
- **Effect size:** — — Safety, tolerability, MTD/RP2D, preliminary clinical activity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BSB-2002 (mutant-NPM1-directed TCR-T)
- **Indication / model:** NPM1-mutated AML, MRD-positive after treatment (2L+); HLA-A*02:01 required + NPM1 mutation type A/D/G/H, MRD-positive
- **Design:** Phase 1 dose-finding, HLA-A*02:01-restricted mutant-NPM1 TCR-T added to standard of care
- **Effect size:** — — Safety, dose
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FH-WT1-E50 (anti-WT1 TCR/CD8ab T-cells) + azacitidine
- **Indication / model:** AML with measurable residual disease (post-therapy) (2L+); HLA-A*02:01, WT1-expressing
- **Design:** Phase 1, autologous HLA-A*02:01-restricted anti-WT1 TCR-T plus azacitidine
- **Effect size:** — — Safety, WT1-directed activity
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** 131I-apamistamab RIT conditioning + allo-HCT
- **Indication / model:** Acute leukemia, radioimmunotherapy conditioning for allo-HCT (2L+); CD45 (targeted conditioning)
- **Design:** Phase 2/3, 131I-apamistamab conditioning + fludarabine/cyclophosphamide/TBI
- **Effect size:** — — Engraftment, remission, survival
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Issa GC/Stein EM (2025) — J Clin Oncol

Reference:PMID 39121437 · **Type:** trial · **Inclusion:** included

- **Intervention:** Revumenib (menin inhibitor)
- **Indication / model:** KMT2A-rearranged relapsed/refractory acute leukemia (2L+); KMT2A rearrangement
- **Design:** Phase 2 single-arm (AUGMENT-101), revumenib monotherapy
- **Effect size:** — — CR / CRh
- **Case fit:** weak
- **Notes:** (toxicity table not extracted; primary endpoint not extracted)

Krakow/Bleakley (2024) — Blood

Reference:PMID 38683966 · **Type:** clinical · **Inclusion:** included

- **Intervention:** HA-1-specific CD8+/CD4+ TCR-T (Bleakley/Fred Hutch platform)
- **Indication / model:** Recurrent leukemia/myeloid neoplasm after allo-HCT (2L+); Recurrent AML/ALL/MDS/JMML after allo-HCT; HLA-A*02:01-positive with recipient HA-1(H)-positive / donor HA-1-negative mismatch. n=9 treated.
- **Design:** Single-arm phase 1 (feasibility/safety)
- **n:** 9
- **Effect size:** 4/9 patients — CR achieved or maintained (4 of 9)
- **Toxicities:** CRS (0/9, 0%); ICANS (0/9, 0%); GVHD (acute) G1-2 (2/9); Neutropenia G>=3 (n=9); Infection G>=3 (n=9)
- **Case fit:** partial
- **Notes:** Per-term AE detail pulled from PMC full text (PMC11406181); the Blood abstract reported only 'no DLTs'. Closest published precedent for the patient's primary HA-1 handle (NCT03326921); commercial HA-1/HA-2 programs TSC-100/101 and BSB-1001 have no published clinical results yet.

Daver/Kantarjian (2024) — Lancet Oncol

Reference:PMID 38423051 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pivekimab sunirine (IMGN632, CD123 ADC)
- **Indication / model:** Relapsed/refractory AML, CD123+ (2L+); CD123+ R/R AML, ECOG 0-1; first-in-human dose-escalation/expansion, RP2D expansion cohort n=29 (whole study n=91).
- **Design:** Phase 1/2 dose-escalation and expansion
- **n:** 91
- **Effect size:** 21 % — ORR at RP2D
- **Variance:** 95% CI 8–40
- **Toxicities:** Febrile neutropenia G>=3 (3/29, 10%); Infusion-related reaction G>=3 (2/29, 7%); Anaemia G>=3 (2/29, 7%); Veno-occlusive disease G>=3 (2/68); Treatment-related death G5 (1/68)
- **Case fit:** partial
- **Notes:** Single-agent CD123 ADC activity in R/R AML matching the patient's CD123 positivity; quantitative antigen density (advisory) is still pending. VOD signal is notable given the planned second transplant.

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BSB-1001 (HA-1-directed TCR-T)
- **Indication / model:** Recurrent AML / ALL / MDS undergoing HLA-matched allo-HCT (2L+); HLA-A*02:01 + HA-1 recipient-positive / donor-negative
- **Design:** Phase 1/2 dose-escalation and expansion of a donor-derived HA-1 TCR-T after allo-HCT
- **Effect size:** — — Safety, dose, anti-leukemic activity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MB-dNPM1-TCR.1 (mutant-NPM1-directed TCR-T)
- **Indication / model:** Relapsed/refractory NPM1-mutated AML (2L+); HLA-A*02:01 required + NPM1 mutation recognized by the TCR
- **Design:** Phase 1/2, autologous HLA-A*02:01-restricted TCR-T targeting the mutant-NPM1 neoantigen
- **Effect size:** — — Safety, dose, response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tagraxofusp + cladribine + cytarabine
- **Indication / model:** CD123-positive relapsed/refractory AML (2L+); CD123+
- **Design:** Phase 1/2, tagraxofusp plus low-intensity cladribine/cytarabine
- **Effect size:** — — Safety, response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) + azacitidine
- **Indication / model:** TP53 Y220C-mutant MDS / AML (2L+); TP53 Y220C
- **Design:** Phase 1b, rezatapopt plus azacitidine
- **Effect size:** — — Safety, response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Garcia-Manero/Sallman (2023) — Lancet Haematol

Reference:PMID 36990622 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246) + venetoclax + azacitidine
- **Indication / model:** TP53-mutant AML, treatment-naive (1L); Treatment-naive TP53-mutant AML with >=1 pathogenic TP53 mutation, ECOG 0-2; triplet expansion evaluable n=39 (whole study n=49). Pan-p53 reactivator, not Y220C-selective.
- **Design:** Phase 1 dose-finding and expansion
- **n:** 49
- **Effect size:** 64 % — Overall response (epren + ven + aza)
- **Variance:** 95% CI 47–79
- **Toxicities:** Febrile neutropenia G>=3 (23/49, 47%); Thrombocytopenia G>=3 (18/49, 37%); Leukopenia G>=3 (12/49, 25%); Anaemia G>=3 (11/49, 22%); Treatment-related death G5 (1/49, 2%)
- **Case fit:** weak
- **Notes:** Eprenetapopt is a pan-p53 reactivator, not Y220C-selective, and the profile flags it low-yield; the front-line treatment-naive setting also differs from this R/R patient. Kept so the board can weigh the pan-reactivator track against the Y220C-selective rezatapopt option.

—/— (2023) — Blood Adv

Reference:PMID 37647601 · **Type:** clinical · **Inclusion:** excluded — Program discontinued, and the phase 1 optimized-dose study enrolled MRD-positive AML, which does not match the patient's overt high-burden relapse (~85% marrow blasts).

- **Intervention:** Vibecotamab (XmAb14045, CD123xCD3 bispecific)
- **Notes:** Excluded: Program discontinued, and the phase 1 optimized-dose study enrolled MRD-positive AML, which does not match the patient's overt high-burden relapse (~85% marrow blasts). — CD123xCD3 bispecific in the same sub-pipeline as flotetuzumab; kept for landscape completeness but not an enrollment route.

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Pivekimab sunirine + FLAG-Ida (fludarabine, cytarabine, idarubicin, G-CSF)
- **Indication / model:** AML / MPAL / MDS (incl. R/R) (2L+); CD123+
- **Design:** Phase 1, pivekimab sunirine added to FLAG-Ida
- **Effect size:** — — Safety, response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Barwe/Gopalakrishnapillai (2022) — J Clin Med

Reference:PMID 35268423 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Flotetuzumab (MGD006) plus cytarabine
- **Indication / model:** Two pediatric AML patient-derived xenograft (PDX) models, CD123+, with infused allogeneic human T-cell effectors
- **Design:** Same CD123xCD3 DART redirection, tested with concurrent cytarabine to see whether chemotherapy

blunts the T-cell-dependent effect.

- **Effect size:** moderate — MGD006 with infused allogeneic T cells drove durable responses in both CD123+ pediatric PDX models and cut peripheral blast burden; adding cytarabine left T-cell proliferation and MGD006 activity intact.
- **Variance:** vs effector T cells without MGD006; MGD006 without cytarabine; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Pediatric AML models needing exogenous human T-cell effectors, so it speaks to mechanism more than to autologous T-cell fitness in a heavily pretreated post-transplant adult.

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TSC-100 (donor-derived HA-1-specific TCR-T)
- **Indication / model:** R/R AML / MDS undergoing HLA-matched allo-HCT (ALLOHA) (2L+); HLA-A*02:01 + HA-1(H) recipient-positive / donor-negative
- **Design:** Open-label phase 1/3, donor-derived HA-1 TCR-T given after HLA-matched allo-HCT, vs standard-of-care control arm
- **Effect size:** — — Safety, donor chimerism, relapse
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TSC-101 (donor-derived HA-2-specific TCR-T)
- **Indication / model:** R/R AML / MDS undergoing HLA-matched allo-HCT (ALLOHA) (2L+); HLA-A*02:01 + HA-2 recipient-positive / donor-negative
- **Design:** Open-label phase 1/3, donor-derived HA-2 TCR-T given after HLA-matched allo-HCT, vs standard-of-care control arm
- **Effect size:** — — Safety, donor chimerism, relapse
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Vibecotamab (XmAb14045, CD123xCD3 bispecific)
- **Indication / model:** MRD-positive AML / MDS (2L+); CD123xCD3 (no biomarker selection)
- **Design:** Phase 2, vibecotamab in MRD-positive disease (terminated)
- **Effect size:** — — MRD conversion
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

Lulla/Leen (2021) — Blood

Reference:PMID 33270816 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Multiantigen leukemia-specific donor T cells (mLST; PRAME/WT1/Survivin/NY-ESO-1)
- **Indication / model:** AML/MDS after allo-HCT (prophylaxis and treatment) (2L+); Post-HCT AML/MDS; 17 at high risk of relapse (prophylaxis) and 8 with active disease; products generated from 29 donors, 25 patients infused.
- **Design:** Single-arm phase 1 dose-escalation
- **n:** 25
- **Effect size:** 2/8 patients (active disease) — Objective response (active-disease cohort: 1 CR, 1 PR)
- **Toxicities:** GVHD (acute) G>=3 (0/25, 0%); GVHD (chronic, extensive) (0/25, 0%)
- **Case fit:** partial
- **Notes:** Published precedent for the multiantigen PRAME/WT1/Survivin donor T-cell approach in AML/MDS (NCT02494167), the same mechanism as the trials.jsonl TAA-T row; WT1 and PRAME expression on the patient's blasts should be confirmed.

Uy/DiPersio (2021) — Blood

Reference:PMID 32929488 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Flotetuzumab (MGD006, CD123xCD3 DART)
- **Indication / model:** Primary induction failure / early-relapse R/R AML (2L+); Primary induction failure or early relapse (<6 mo) R/R AML; benefit enriched in an immune-infiltrated tumor microenvironment. RP2D PIF/ER subset n=30 (whole study n=88).
- **Design:** Multicenter open-label phase 1/2
- **n:** 88
- **Effect size:** 26.7 % — CR/CRh (PIF/ER at RP2D)
- **Toxicities:** CRS / infusion-related reaction (n=88); CRS / infusion-related reaction G>=3 (n=88)
- **Case fit:** partial
- **Notes:** Efficacy precedent for CD123xCD3 bispecific salvage in chemo-refractory AML close to the patient's state; MacroGenics has since deprioritized the program, so it is mechanism evidence rather than an open route. Abstract gives no per-term grade-3 CRS rate; CT.gov NCT02152956 AE tab holds the breakdown.

Sallman/Komrokji (2021) — J Clin Oncol

Reference:PMID 33449813 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246) + azacitidine
- **Indication / model:** TP53-mutant MDS or oligoblastic AML (1L); TP53-mutant MDS (n=40), oligoblastic AML with 20-30% blasts (n=11), and MDS/MPN (n=4); frontline. 35% went on to allo-HCT.
- **Design:** Phase 1b/2 single-arm
- **n:** 55
- **Effect size:** 71 % — Overall response (epren + aza)
- **Toxicities:** Febrile neutropenia G>=3 (n=55, 33%); Leukopenia G>=3 (n=55, 29%); Neutropenia G>=3 (n=55, 29%)

- **Case fit:** weak
- **Notes:** The doublet phase 1b/2 in TP53-mutant MDS/oligoblastic AML; the subsequent randomized phase 3 in TP53-mutant MDS did not meet its primary endpoint, and eprenetapopt is not Y220C-selective, reinforcing the low-yield framing for this patient.

—/— (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NTLA-5001 (WT1-directed TCR-T)
- **Indication / model:** Acute myeloid leukemia (2L+); HLA-A*02:01, WT1+
- **Design:** Phase 1/2 CRISPR-engineered WT1 TCR-T (terminated)
- **Effect size:** — — Safety, dose (program halted)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Vadakekolathu/Rutella (2020) — Blood Adv

Reference:PMID 33057635 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Flotetuzumab (MGD006) in TP53-abnormal AML
- **Indication / model:** Primary AML bone-marrow samples profiled by NanoString immune panel plus BM from flotetuzumab-treated R/R AML
- **Design:** TP53-mutated or 17p-deleted AML carries an inflamed marrow immune landscape (higher IFNG, FOXP3, immune-checkpoint and senescence markers), which the authors reasoned would sensitize to CD123xCD3 redirection.
- **n:** n=64 primary BM (42 TP53-mut, 22 TP53-WT); 45 flotetuzumab-treated, 15 with TP53 abnormalities
- **Effect size:** TP53-abnormal AML showed a more inflamed, IFN-gamma-high microenvironment than TP53-WT. Among R/R AML with TP53 abnormalities on the flotetuzumab trial, 7 of 15 reached under 5% marrow blasts, and responders had higher baseline tumor-inflammation, CD8 and PD1 scores.
- **Variance:** vs TP53-wild-type AML samples; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Correlative biomarker study whose response numbers come from the flotetuzumab trial cohort, so it is predictive rationale rather than a preclinical efficacy model. Bears directly on this patient's TP53-aberrant status if confirmed.

Orozco/Pagel (2020) — Mol Cancer Ther

Reference:PMID 33082277 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Anti-CD45 (BC8) radioimmunotherapy, bispecific 90Y-DOTA construct
- **Indication / model:** HEL human leukemia xenografts (anti-human CD45 BC8 construct); syngeneic disseminated SJL murine leukemia (anti-mouse 30F11)
- **Design:** A bispecific anti-CD45 x anti-Y-DOTA construct (BC8 for human CD45, the apamistamab parent antibody) pretargets leukemic and marrow cells, then captures a separately injected 90Y-DOTA radioligand to concentrate beta-radiation at CD45+ hematolymphoid sites.

- **Effect size:** strong — 70% of mice bearing HEL human leukemia xenografts survived past 170 days after BC8-Fc-C825 plus 90Y-DOTA-biotin, while untreated controls needed euthanasia by day 32. The syngeneic SJL model showed high spleen/marrow uptake and a survival gain (median 43 vs 30 days).
- **Variance:** vs untreated leukemic mice; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Pretargeted bispecific format with 90Y, not the directly-131I-labeled apamistamab (Iomab-B) used clinically; it shares the BC8 anti-CD45 targeting and the marrow-directed-radiation rationale for transplant conditioning.

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) +/- pembrolizumab
- **Indication / model:** Advanced solid tumors, TP53 Y220C (2L+); TP53 Y220C
- **Design:** Phase 1/2 (PYNNACLE), rezatapopt monotherapy and with pembrolizumab
- **Effect size:** — — Objective response, safety
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Chapuis/Greenberg (2019) — Nat Med

Reference:PMID 31235963 · **Type:** clinical · **Inclusion:** included

- **Intervention:** WT1-TCRC4 EBV-specific CD8+ Tcm (Chapuis/Greenberg)
- **Indication / model:** High-risk AML, post-HCT relapse prophylaxis (maintenance); HLA-A*02:01-positive, WT1-positive high-risk AML given as post-HCT relapse prophylaxis; compared with a concurrent 88-patient similar-risk cohort.
- **Design:** Phase 1/2 single-arm with concurrent comparator cohort
- **n:** 12
- **Effect size:** 100 % at median 44 mo — Relapse-free survival (prophylaxis)
- **Variance:** p=0.002
- **Toxicities:** GVHD (n=12)
- **Case fit:** weak
- **Notes:** Anchors the WT1 A*02:01 TCR-T arm; a prophylaxis product since succeeded by FH-WT1-E50, so precedent rather than an open slot. Abstract carries no AE table; full text (PMC6982533) would be needed for per-term safety.

Pemmaraju/Konopleva (2019) — N Engl J Med

Reference:PMID 31018069 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Tagraxofusp (SL-401, IL-3/diphtheria-toxin fusion)
- **Indication / model:** Blastic plasmacytoid dendritic-cell neoplasm (BPDCN), CD123+ (any); CD123+ BPDCN; 29 previously untreated at 12 μg/kg (primary-outcome cohort) plus 15 previously treated. Not the patient's disease.

- **Design:** Open-label single-arm phase 2 (multicohort)
- **n:** 47
- **Effect size:** 72 % — CR/clinical CR (first-line 12 µg/kg)
- **Toxicities:** Capillary leak syndrome (n=47, 19%); ALT increased (n=47, 64%); AST increased (n=47, 60%); Hypoalbuminemia (n=47, 55%); Peripheral edema (n=47, 51%); Thrombocytopenia (n=47, 49%)
- **Case fit:** cross_tumor_only
- **Notes:** Registration study behind the CD123-directed BPDCN approval; carries the mechanism and off-label precedent for the patient's CD123 handle, but BPDCN is not AML so this is cross-tumor read-across. Capillary-leak and hepatic risk are relevant given the undocumented organ function.

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CD123-CD33 cCAR T cells
- **Indication / model:** Relapsed/refractory high-risk AML / MDS (2L+); CD123+ and/or CD33+
- **Design:** Early phase 1 compound CAR-T co-targeting CD123 and CD33
- **Effect size:** — — Safety, dose
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CLL-1/CD33/CD123-directed CAR-T
- **Indication / model:** Relapsed/refractory AML (2L+); CLL-1 and/or CD33 and/or CD123
- **Design:** Phase 1/2 multi-target CAR-T against CLL-1, CD33, CD123
- **Effect size:** — — Safety, response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Ziftomenib (menin inhibitor)
- **Indication / model:** Relapsed/refractory AML (NPM1-mutant or KMT2A-rearranged) (2L+); NPM1-mutant / KMT2A rearrangement
- **Design:** Phase 1/2 first-in-human (COMET-001), ziftomenib monotherapy
- **Effect size:** — — Response, safety
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Kovtun/Chittenden (2018) — Blood Adv

Reference:PMID 29661755 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Pivekimab sunirine (IMGN632, CD123 ADC)
- **Indication / model:** AML cell lines, patient-derived AML samples, and multiple AML xenograft models (mouse)
- **Design:** IMGN632 links a humanized anti-CD123 antibody to a DNA mono-alkylating indolinobenzodiazepine (IGN) payload; CD123 sits at higher density on leukemic blasts and stem cells than on normal hematopoietic progenitors, which opens the therapeutic window.
- **Effect size:** strong — Low-picomolar killing across AML lines and patient samples irrespective of multidrug resistance or disease status, at doses below those affecting normal myeloid progenitors, with the mono-alkylator payload over 40-fold less toxic to normal progenitors than the crosslinker version. Single-agent activity carried through several AML xenograft models.
- **Variance:** vs X-ADC (crosslinking-IGN version of the same G4723A antibody); non-binding controls; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** ImmunoGen sponsor study. Potency tracks antigen density, so the patient's pending quantitative CD123 read bears directly on read-across.

Mani/Muthusamy (2018) — Haematologica

Reference:PMID 29773600 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Tagraxofusp (SL-401) against CD34+CD123+ AML/MDS cells
- **Indication / model:** Primary CD34+CD123+ cells from AML and MDS patients; AML patient-derived xenograft mice; normal cord-blood/marrow progenitors
- **Design:** SL-401 (tagraxofusp) directs the diphtheria-toxin catalytic domain to CD123 on leukemic and MDS stem/progenitor cells, where the internalized toxin blocks protein synthesis, reaching even low-CD123 cells.
- **Effect size:** moderate — SL-401 killed CD123+ AML and MDS primary blasts and CD34+CD123+ cells, including those with low CD123 density, while sparing CD123-negative lymphoid cells, and prolonged survival in AML PDX mice. Normal cord-blood and marrow progenitors were also hit, an on-target off-leukemia effect.
- **Variance:** vs CD123-negative lymphoid cells; healthy-donor progenitors; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Killing of normal CD34+CD123+ progenitors flags myelosuppression; the authors frame SL-401 as a bridge-to-transplant, matching this patient's second-transplant intent. Stemline-affiliated co-author.

Dossa/Bleakley (2018) — Blood

Reference:PMID 29051183 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HA-1-specific CD8+/CD4+ TCR-T (Bleakley/Fred Hutch platform)
- **Indication / model:** Primary human leukemia lysis by lentivirally transduced CD4+ and CD8+ T cells (in-vitro functional characterization)
- **Design:** A four-part transgene carries the HA-1 minor-histocompatibility-antigen TCR, a CD8 coreceptor so the class-I-restricted TCR works in CD4+ cells, an inducible caspase-9 safety switch, and a CD34-CD20 selection/tracking epitope; HA-1's hematopoietic-restricted expression confines the graft-versus-leukemia effect to the blood compartment.
- **Effect size:** moderate — CD4+ and CD8+ T cells engineered with the HA-1 TCR transgene lysed primary

leukemia and mounted antigen-specific functional responses. The memory-only, CD8-coreceptor design was built to hold down alloreactivity from the co-expressed native TCR.
- **Variance:** vs untransduced / non-HA-1-specific T cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Functional in-vitro characterization of the construct with no in-vivo tumor-control arm in this report. Requires HLA-A*02:01 plus recipient HA-1(H)-positive / donor HA-1-negative mismatch, both pending here.

Campagne/Mager (2018) — Clin Cancer Res

Reference:PMID 29463552 · **Type:** preclinical · **Inclusion:** excluded — PK/PD and immunogenicity modeling of the CD3xCD123 DART in nonhuman primates; the efficacy and safety proof-of-concept for this mechanism is already carried by Chichili 2015, so it adds pharmacometric detail rather than new target-directed evidence.

- **Intervention:** CD3xCD123 DART PK/PD model (nonhuman primates)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: PK/PD and immunogenicity modeling of the CD3xCD123 DART in nonhuman primates; the efficacy and safety proof-of-concept for this mechanism is already carried by Chichili 2015, so it adds pharmacometric detail rather than new target-directed evidence.

Tawara/Shiku (2017) — Blood

Reference:PMID 28860210 · **Type:** clinical · **Inclusion:** included

- **Intervention:** WT1-siTCR gene-transduced lymphocytes (TBI-1301)
- **Indication / model:** Refractory AML and high-risk MDS (2L+); HLA-A*24:02-positive, WT1-positive refractory AML and high-risk MDS; two dose cohorts, n=8.
- **Design:** First-in-human phase 1 (safety, cell kinetics)
- **n:** 8
- **Effect size:** 2/8 patients with transient blast decrease — Transient marrow blast reduction; T-cell persistence
- **Toxicities:** On-target off-tumor (normal tissue) toxicity (0/8, 0%)
- **Case fit:** weak
- **Notes:** The A*24:02-restricted WT1-siTCR fallback lever if the patient types HLA-A*02:01-negative. Small early-phase Japanese cohort; abstract reports no per-term AE table beyond the absence of normal-tissue toxicity.

—/— (2016)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BPX-701 (PRAME TCR-T with iCasp9 safety switch) + rimiducid
- **Indication / model:** Acute myeloid leukemia / MDS (2L+); HLA-A*02:01, PRAME+
- **Design:** Phase 1/2 rimiducid-controllable PRAME TCR-T (terminated)
- **Effect size:** — — Safety, dose (program halted)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Chichili/Bonvini (2015) — Sci Transl Med

Reference:PMID 26019218 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Flotetuzumab (MGD006, CD123xCD3 DART)
- **Indication / model:** KG-1a AML xenograft in PBMC-reconstituted NSG/beta2m-/- mice; cynomolgus macaques
- **Design:** MGD006 co-engages CD3 on T cells and CD123 on blasts through a dual-affinity re-targeting format, redirecting host T cells to lyse CD123+ cells alongside T-cell activation and expansion; a short half-life dictates continuous infusion.
- **Effect size:** strong — Continuous low-dose MGD006 cleared engrafted KG-1a cells in humanized mice at 0.5 ug/kg/day. In macaques it depleted circulating CD123+ cells within 72 h and held that depletion across 4 weeks, with cytokine release easing after the first dose and no neutropenia or thrombocytopenia.
- **Variance:** vs untreated / pre-infusion baseline; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** MacroGenics study; the clinical program was later deprioritized. KG-1a is CD123+ but not MDS-related or TP53-defined.

Hills/Burnett (2014) — Lancet Oncol

Reference:PMID 25008258 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin added to induction (IPD meta-analysis)
- **Indication / model:** Adult AML (non-APL), front-line induction (1L); 3325 adults with AML (excluding APL) in five randomized induction trials; cytogenetic risk subgroups analyzed, including an adverse-cytogenetics stratum.
- **Design:** Individual-patient-data meta-analysis of RCTs
- **n:** 3325
- **Effect size:** 0.9 OR — 5-year overall survival (addition of GO)
- **Variance:** 95% CI 0.82–0.98; p=0.01
- **Toxicities:** Early death G5 (n=3325)
- **Case fit:** weak
- **Notes:** Highest-tier CD33 evidence, and the decision-relevant point for this patient is the null adverse-cytogenetics subgroup (OR 0.99): patients with adverse karyotype derived no survival benefit, which directly tempers the CD33 handle here given the complex/TP53-aberrant karyotype.

—/— (2014)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tumor-associated-antigen-specific T cells (TAA-T; PRAME/WT1/Survivin)
- **Indication / model:** Very-high-risk hematologic malignancies (AML/ALL/MDS), largely post-allo-HCT (2L+); WT1 / PRAME / Survivin expression (multi-tumor-associated-antigen)
- **Design:** Phase 1, donor- or patient-derived multi-antigen-specific cytotoxic T lymphocytes (TAA-T) primed against PRAME, WT1 and Survivin
- **Effect size:** — — Feasibility, safety, anti-leukemic activity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

Castaigne/Dombret (2012) — Lancet

Reference:PMID 22482940 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin added to induction chemotherapy (ALFA-0701)
- **Indication / model:** De novo AML, front-line (1L); Previously untreated de novo AML age 50-70 (excludes APL); CD33 expression not required for entry. 280 randomized.
- **Design:** Randomized open-label phase 3
- **n:** 280
- **Effect size:** 0.58 HR — Event-free survival (GO + chemo vs chemo)
- **Variance:** 95% CI 0.43–0.78; p=0.0003
- **Toxicities:** Persistent thrombocytopenia (22/139, 16%); Death from toxicity G5 (n=139)
- **Case fit:** weak
- **Notes:** Pivotal RCT behind the fractionated-dose gemtuzumab schedule; front-line de novo add-on, so line and setting differ from this R/R patient. Benefit was concentrated in favourable/intermediate cytogenetics.

Ochi/Yasukawa (2011) — Blood

Reference:PMID 21673345 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** WT1-siTCR gene-transduced lymphocytes (TBI-1301)
- **Indication / model:** WT1-siTCR-transduced patient T cells lysing autologous leukemia; mouse leukemia xenograft (adoptive transfer)
- **Design:** The siTCR vector encodes the WT1-specific TCR plus siRNAs that silence endogenous TCR chains, cutting mispairing and raising surface expression of the introduced TCR.
- **Effect size:** moderate — WT1-siTCR T cells outperformed conventional WT1-TCR cells on antileukemia reactivity, expanded well on repeat WT1-peptide stimulation, and lysed autologous leukemia while sparing normal hematopoietic progenitors. Transferred cells controlled leukemia in a xenograft model without suppressing human hematopoiesis.
- **Variance:** vs conventional codon-optimized WT1-TCR (no siRNA) T cells; normal hematopoietic progenitors; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Relevant only on the HLA-A*24:02 branch (the fallback if the patient is A*02:01-negative); WT1 expression on the patient's blasts is not yet quantified.

Pagel/Press (2011) — Blood

Reference:PMID 21613259 · **Type:** preclinical · **Inclusion:** excluded — Anti-CD45 streptavidin-pretargeting predecessor using 213Bi in a human myeloid-leukemia xenograft; a strong POC, but the endogenous-biotin and immunogenicity limits of the streptavidin approach were later addressed by the BC8 bispecific construct (Orozco 2020), which better represents the apamistamab targeting used clinically.

- **Intervention:** Anti-CD45 pretargeted radioimmunotherapy, 213Bi (streptavidin platform)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Anti-CD45 streptavidin-pretargeting predecessor using 213Bi in a human myeloid-leukemia

xenograft; a strong POC, but the endogenous-biotin and immunogenicity limits of the streptavidin approach were later addressed by the BC8 bispecific construct (Orozco 2020), which better represents the apamistamab targeting used clinically.

Lambert/Bykov (2009) — Cancer Cell

Reference:PMID 19411067 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246) p53 reactivation mechanism
- **Indication / model:** Mutant-p53 human tumor cell lines; biochemical thiol-adduct binding assays
- **Design:** PRIMA-1, the parent of APR-246/eprenetapopt, is converted to a reactive methylene-quinuclidinone that forms covalent adducts with thiol groups in the mutant p53 core domain; that covalent modification alone can restore a pro-apoptotic conformation.
- **Effect size:** PRIMA-1-derived compounds bound covalently to cysteine thiols in mutant p53, and covalent modification by itself triggered apoptosis in mutant-p53 tumor cells. The work pinned down the covalent-adduct mechanism that APR-246 shares.
- **Variance:** translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** A pan-mutant mechanism, not Y220C-selective; the profile flags eprenetapopt as low-yield and the randomized TP53-mutant MDS phase 3 missed its endpoint. The assays used PRIMA-1, the parent of clinical APR-246.

Black/Frankel (2003) — Leukemia

Reference:PMID 12529673 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Tagraxofusp (DT388IL3, IL-3/diphtheria-toxin fusion)
- **Indication / model:** SCID mice inoculated i.v./i.p./s.c. with TF1 (v-SRC-transfected) human AML cells
- **Design:** DT388IL3 fuses the diphtheria-toxin catalytic and translocation domains to IL-3, delivering the toxin to CD123 (IL-3Ralpha)-expressing blasts where it halts protein synthesis.
- **n:** n=20-49 mice per inoculation route
- **Effect size:** strong — Only 5 of 49 i.v.-inoculated animals died with leukemia after DT388IL3, and median disease-free survival stretched beyond 120 days versus 37 days for controls. The gain held across i.p. and s.c. models, while the IL-2 fusion DAB389IL2 gave none.
- **Variance:** vs saline; comparators DT388GMCSF, DAB389IL2, cytarabine, anti-CD33-calicheamicin; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** TF1/v-SRC is an engineered differentiated-AML model, not MDS-related AML; tagraxofusp's registration is in BPDCN, so AML use is off-label read-across.

Hamann/Bernstein (2002) — Bioconjug Chem

Reference:PMID 11792178 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin (CD33 ADC)
- **Indication / model:** HL-60 CD33+ leukemia cells in culture and HL-60 xenografts in mice; AML patient marrow

colony-forming assays

- **Design:** Gemtuzumab ozogamicin joins the humanized anti-CD33 antibody to a calicheamicin derivative through an acid-labile hydrazone; CD33 internalization frees the DNA-cleaving payload inside the blast while sparing CD33-negative stem cells.
- **Effect size:** strong — Sub-picogram IC50 against HL-60 in culture, and doses as low as 50 ug cal/kg given three times produced long-term tumor-free survival in HL-60 xenografts while a non-binding conjugate stayed largely inactive. Low-nanogram concentrations selectively inhibited AML colony formation from a subset of patient marrows.
- **Variance:** vs non-binding calicheamicin conjugate; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** HL-60 is a CD33+ AML line without the patient's adverse-karyotype, TP53, or MDS-related biology; the Hills 2014 IPD meta-analysis found no survival benefit in the adverse-cytogenetics subgroup, tempering read-across.

Sievers/Mylotarg Study Group (2001) — J Clin Oncol

Reference:PMID 11432892 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin (CD33 ADC)
- **Indication / model:** CD33+ AML in untreated first relapse (2L+); CD33+ AML in untreated first relapse, no antecedent hematologic disorder; median age 61. Pooled across three open-label trials.
- **Design:** Single-arm phase 2 (three pooled open-label trials)
- **n:** 142
- **Effect size:** 30 % — Overall remission (CR + CRp)
- **Toxicities:** Hyperbilirubinemia G>=3 (n=142, 23%); Transaminase elevation G>=3 (n=142, 17%); Infection (n=142, 28%); Mucositis G>=3 (n=142, 4%); Nausea/vomiting (severe) G>=3 (n=142, 11%)
- **Case fit:** partial
- **Notes:** On-label CD33 ADC monotherapy in first relapse; single-agent depth is modest (30%) and the patient is further along with FLAG-IDA/venetoclax failure and extramedullary disease. Hepatic/VOD risk matters ahead of a second transplant.